

The $d = 4$ fiber of the graded order law: a reduction to Schur support, and the obstruction

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Abstract

Let $G_\lambda(q) = \sum_{T \in \text{SYT}(\lambda)} q^{s(T)}$ be the graded branch-exponent generating function, where $s(T) = \sum_{i \in \text{Des}(T)} w_i$ with the parity weights $w_i = 2i - 1$ if $n - i$ is odd and $w_i = 0$ otherwise. At the cube root of unity the fiber value $G_\lambda(\zeta_3)$ vanishes exactly on the two 3-cores $(3, 1), (2, 1, 1)$, by a Schur-positivity argument [Clio 2026-06-02]. At the *fourth* root of unity $\zeta_4 = i$ the value is genuinely complex and that argument cannot transplant. We prove instead a structural *reduction*: because the active descent set is an arithmetic progression of common difference 2, at ζ_4 *all* of its factors in the master identity coincide, so that for $n = 2m$ even

$$G_\lambda(i) = \langle s_\lambda, \psi^m \rangle, \quad \psi := h_2 + i e_2 = s_2 + i s_{11},$$

a single pure power, with the odd- n analogue $G_\lambda(i) = \langle s_\lambda, h_1 \bar{\psi}^m \rangle$. Consequently $G_\lambda(i) = 0$ if and only if λ lies outside the (complex) Schur support of ψ^m . We record the equivalent Hermite-determinant and Gaussian-integral forms, prove the boundary cases, verify computationally that $(2, 2)$ is the unique vanisher for $|\lambda| \leq 16$, and isolate precisely the remaining statement:

$\text{supp}_{\mathbb{C}}(\psi^m) \text{ is full for } m \geq 3, \text{ and } \text{supp}_{\mathbb{C}}(\psi^2) = \{\lambda \vdash 4\} \setminus \{(2, 2)\}.$

The $m = 2$ clause is proved by a one-line expansion. The $m \geq 3$ clause is the target theorem; we explain why it is governed by a ζ_4 cancellation (a 4-quotient phenomenon) rather than by positivity, and leave it open with a precise route.

1 Setup

Throughout $\lambda \vdash n$, and for $T \in \text{SYT}(\lambda)$, $\text{Des}(T) = \{i \in [n-1] : i+1 \text{ lies in a strictly lower row than } i\}$. Fix

$$A = \{k \in [n-1] : n - k \text{ odd}\}, \quad w_k = (2k - 1) [k \in A], \quad s(T) = \sum_{i \in \text{Des}(T)} w_i,$$

and set $m = |A| = \lfloor n/2 \rfloor$, $e = n \bmod 2$. We work in the ring Λ of symmetric functions with the Hall inner product, for which $\langle s_\lambda, h_\mu \rangle = K_{\lambda\mu}$ and $\langle s_\lambda, p_\mu \rangle = \chi^\lambda(\mu)$. We use $h_2 = \frac{1}{2}(p_1^2 + p_2)$, $e_2 = \frac{1}{2}(p_1^2 - p_2)$, so $h_2 + e_2 = p_1^2$ and $h_2 - e_2 = p_2$. For $f \in \Lambda$ write $\text{supp}_{\mathbb{C}}(f) = \{\mu : \langle s_\mu, f \rangle \neq 0\}$.

We recall the master identity, proved (for any ζ) in the ζ_3 note.

Proposition 1 (Master identity). *For any $\zeta \in \mathbb{C}$, $G_\lambda(\zeta) = \left\langle s_\lambda, h_1^e \prod_{k \in A} (h_2 + e_2 \zeta^{2k-1}) \right\rangle$.*

2 The reduction: a single pure power at ζ_4

This is the structural phenomenon that distinguishes $d = 4$ from general d .

Lemma 1 (Coincidence of factors). *At $\zeta = i$, for every $k \in A$ the factor $h_2 + e_2 i^{2k-1}$ equals $h_2 + i e_2$ if n is even, and $h_2 - i e_2$ if n is odd. In particular all $|A| = m$ factors are equal.*

Proof. Since $2k - 1$ is odd, $i^{2k-1} = i$ if $2k - 1 \equiv 1 \pmod{4}$, i.e. k odd, and $i^{2k-1} = i^3 = -i$ if k even. The active set A is an arithmetic progression of common difference 2: $A = \{1, 3, \dots, n-1\}$ when n is even (all k odd) and $A = \{2, 4, \dots, n-1\}$ when n is odd (all k even). Hence within A the parity of k is constant, and i^{2k-1} takes a single value: i for n even, $-i$ for n odd. $\square$

Theorem 1 (Reduction). *Write $\psi = h_2 + i e_2$ and $\bar{\psi} = h_2 - i e_2$. Then*

$$n = 2m \text{ even: } G_\lambda(i) = \langle s_\lambda, \psi^m \rangle, \quad n = 2m + 1 \text{ odd: } G_\lambda(i) = \langle s_\lambda, h_1 \bar{\psi}^m \rangle.$$

Consequently, for n even, $G_\lambda(i) = 0$ if and only if $\lambda \notin \text{supp}_{\mathbb{C}}(\psi^m)$.

Proof. Immediate from Proposition 1 and Lemma 1: the product $\prod_{k \in A} (h_2 + e_2 i^{2k-1})$ collapses to ψ^m (resp. $\bar{\psi}^m$), and $h_1^e = 1$ (resp. h_1). The support statement is the definition of $\langle s_\lambda, \psi^m \rangle \neq 0$. $\square$

Remark 1. This is exactly the point where the ζ_3 proof's strategy breaks. At ζ_3 the factors split into a conjugate pair $\phi, \bar{\phi}$ whose product $\Theta = \phi \bar{\phi} = s_4 + 2s_{22} + s_{14}$ is *Schur-positive*, turning the fiber value into a manifestly nonnegative integer. At ζ_4 with n even *all* factors coincide as ψ ; there is no conjugate partner inside the product, so ψ^m has genuinely complex coefficients and no positivity is available. The value $G_\lambda(i)$ is a Gaussian integer that can a priori vanish by cancellation.

We henceforth treat the even case $n = 2m$ and write $P_\lambda := \langle s_\lambda, \psi^m \rangle = G_\lambda(i) \in \mathbb{Z}[i]$. The odd case is recovered from $G_\lambda(i) = \langle s_\lambda, h_1 \bar{\psi}^m \rangle = \sum_\mu \overline{P_\mu}$, the sum over the even partitions μ obtained by deleting one corner box of λ .

3 Three faces of P_λ

3.1 The interpolating polynomial and the skewing recursion

Let $P_\lambda(t) := \langle s_\lambda, (h_2 + t e_2)^m \rangle \in \mathbb{Z}_{\geq 0}[t]$, so $P_\lambda = P_\lambda(i)$. Because $\langle s_\lambda, (h_2 + t e_2)^m \rangle$ is a product of the Schur-positive functions h_2, e_2 , its coefficients are nonnegative integers. The skewing operators $h_2^\perp, e_2^\perp$ remove horizontal, resp. vertical, 2-strips, giving the recursion

$$P_\lambda(t) = \sum_{\lambda/\mu \text{ horiz. 2-strip}} P_\mu(t) + t \sum_{\lambda/\mu \text{ vert. 2-strip}} P_\mu(t), \quad P_\emptyset = 1. \quad (1)$$

Three evaluations are pinned down: $P_\lambda(1) = \langle s_\lambda, p_1^{2m} \rangle = f^\lambda$; $P_\lambda(-1) = \langle s_\lambda, p_2^m \rangle = \chi^\lambda(2^m) = \pm \binom{m}{|\lambda^0|} f^{\lambda^0} f^{\lambda^1}$ (the signed number of standard domino tableaux, nonzero iff λ has empty 2-core); and $P_\lambda(0) = K_{\lambda, (2^m)}$. Conjugation acts by coefficient reversal,

$$P_{\lambda'}(t) = t^m P_\lambda(1/t), \quad \text{hence} \quad P_{\lambda'}(i) = i^m \overline{P_\lambda(i)}, \quad (2)$$

using $\omega(h_2 + t e_2) = t(h_2 + t^{-1} e_2)$. In particular $G_\lambda(i) = 0 \iff G_{\lambda'}(i) = 0$, and the self-conjugate $(2, 2)$ is consistent with this.

3.2 Hermite determinant and Gaussian integral

Setting $p_{\geq 3} = 0$ is the ring homomorphism $\Lambda \rightarrow \mathbb{C}[A, B]$, $p_1 \mapsto A$, $p_2 \mapsto B$. Under it $h_r \mapsto [z^r] \exp(Az + \frac{B}{2}z^2)$, an Appell/Hermite polynomial, so by Jacobi–Trudi $s_\lambda(p_1=A, p_2=B, 0) = \det(h_{\lambda_i-i+j}(A, B, 0))$ is a Hermite determinant. A Hubbard–Stratonovich linearisation of p_1^2 then yields the integral representation

$$P_\lambda(t) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-s^2} s_\lambda(p_1=2s, p_2=2t, p_{\geq 3}=0) ds, \quad (3)$$

equivalently $G_\lambda(i) = \mathbb{E}_{Y \sim N(0,2)} [s_\lambda(p_1=Y, p_2=-2i, 0)]$ up to the unit $\beta^m i^m$, $\beta = \frac{1-i}{2}$. (Verified numerically for all λ with $n \leq 10$.)

3.3 A Parseval identity

By orthonormality of Schur functions,

$$\sum_{\lambda \vdash 2m} |G_\lambda(i)|^2 = \langle \psi^m, \bar{\psi}^m \rangle = 2^{-m} \sum_{a=0}^m \binom{m}{a}^2 2^a a! (2m-2a)! > 0, \quad (4)$$

since $\langle p_\mu, p_\nu \rangle = z_\mu \delta_{\mu\nu}$ and $\alpha\beta = \frac{1}{2}$ for $\psi = \alpha p_1^2 + \beta p_2$ ($\alpha = \frac{1+i}{2}$). (Verified for $m \leq 5$: the totals are 2, 12, 168, 4512, 198720.) This shows the fiber values are not all zero but does not by itself isolate $(2, 2)$.

4 What is proved, and the boundary

Proposition 2 (Extreme shapes). $P_{(2m)}(i) = 1$ and $P_{(1^{2m})}(i) = i^m$; both are nonzero.

Proof. A one-row shape admits only horizontal 2-strip removals, and there is a unique chain $(2m) \rightarrow (2m-2) \rightarrow \cdots \rightarrow \emptyset$; by (1) its weight is t^0 , so $P_{(2m)}(t) = 1$. Dually $P_{(1^{2m})}(t) = t^m$ by (2). $\square$

Proposition 3 (Kostka expansion and a Gaussian-prime obstruction). Since $e_2 = h_1^2 - h_2$, one has $\psi = i h_1^2 + (1-i)h_2$, hence

$$\psi^m = \sum_{k=0}^m \binom{m}{k} i^{m-k} (1-i)^k h_{(2^k 1^{2m-2k})}, \quad \text{so} \quad G_\lambda(i) = i^m \sum_{k=0}^m \binom{m}{k} (-1-i)^k K_{\lambda, (2^k 1^{2m-2k})}.$$

Let $(1+i)$ be the Gaussian prime above 2. Reducing modulo $(1+i)$ kills every $k \geq 1$ term, leaving $G_\lambda(i) \equiv i^m K_{\lambda, (1^{2m})} = i^m f^\lambda \pmod{(1+i)}$. Therefore

$$f^\lambda \text{ odd} \implies G_\lambda(i) \neq 0.$$

In particular every vanisher has f^λ even; consistent with $f^{(2,2)} = 2$. (Both displays verified for all λ with $n \leq 12$, 0 exceptions; the sufficient condition has no exception through $n = 14$.)

Proof. The expansion is the binomial theorem applied to $\psi = i h_1^2 + (1-i)h_2$ together with $h_1^{2(m-k)} h_2^k = h_{(2^k 1^{2m-2k})}$; pairing with s_λ uses $\langle s_\lambda, h_\mu \rangle = K_{\lambda\mu}$ and $i^{m-k} (1-i)^k = i^m ((1-i)/i)^k = i^m (-1-i)^k$. Since $(-1-i) = -(1+i)$, each $k \geq 1$ summand lies in $(1+i)\mathbb{Z}[i]$, so $G_\lambda(i) \equiv i^m f^\lambda \pmod{(1+i)}$. As $(1+i) \mid f^\lambda$ in $\mathbb{Z}[i]$ iff $2 \mid f^\lambda$, an odd f^λ forces $G_\lambda(i) \not\equiv 0$, hence $G_\lambda(i) \neq 0$. $\square$

Remark 2. The obstruction is sharp at the level of one prime: it cannot see vanishers, which all have f^λ even. One might hope to iterate $(1+i)$ -adically, but the even- and odd- k partial sums do *not* separately vanish at a vanisher (for $(2,2)$ they are $2+2i$ and $-2-2i$, which cancel), so a single congruence does not suffice; controlling $v_{(1+i)}(G_\lambda(i))$ in general appears to require the 2-adic valuations of the Kostka numbers $K_{\lambda, (2^k 1^{2m-2k})}$ (the 2-core tower).

Proposition 4 (The $m = 2$ support, exactly). $\psi^2 = s_4 + (1+2i)s_{31} + (2i-1)s_{211} - s_{14}$. Hence $\text{supp}_{\mathbb{C}}(\psi^2) = \{(4), (3,1), (2,1,1), (1^4)\} = \{\lambda \vdash 4\} \setminus \{(2,2)\}$, and $(2,2)$ is the unique partition of 4 with $G_\lambda(i) = 0$.

Proof. With $h_2 = s_2$, $e_2 = s_{11}$ and Pieri, $s_2^2 = s_4 + s_{31} + s_{22}$, $s_{11}^2 = s_{22} + s_{211} + s_{14}$, $s_2 s_{11} = s_{31} + s_{211}$. Therefore $\psi^2 = (s_2 + i s_{11})^2 = s_2^2 + 2i s_2 s_{11} - s_{11}^2 = s_4 + (1+2i)s_{31} + (1-1)s_{22} + (2i-1)s_{211} - s_{14}$. The coefficient of s_{22} is $1-1=0$: the real parts coming from s_2^2 and $-s_{11}^2$ cancel, and the imaginary cross term $2i s_2 s_{11}$ contributes nothing to s_{22} . Equivalently $G_{(2,2)}(i) = \chi^{(2,2)}(1^4)\alpha^2 + \chi^{(2,2)}(2,2)\beta^2 = 2(\alpha^2 + \beta^2) = 2 \cdot 0 = 0$ since $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 1 - 1 = 0$. $\square$

Theorem 2 (Computational verification). *For every λ with $|\lambda| \leq 16$ (both parities; 913 shapes through $n = 16$), $G_\lambda(i) = 0$ if and only if $\lambda = (2,2)$.*

5 The remaining statement and why it is hard

By Theorem 1 the full theorem $G_\lambda(i) = 0 \iff \lambda = (2,2)$ is *equivalent* to the following.

Conjecture 1 (Full support). *For $m \geq 3$, $\text{supp}_{\mathbb{C}}(\psi^m) = \{\lambda \vdash 2m\}$; and for $m = 2$ the support omits exactly $(2,2)$. Equivalently, $P_\lambda(i) \neq 0$ for all $\lambda \neq (2,2)$.*

The $m = 2$ clause is Proposition 4; the $m = 1$ case is $\psi = s_2 + i s_{11}$ (full). What remains is the $m \geq 3$ clause.

Why positivity fails. Writing $P_\lambda(i) = \text{Re } P_\lambda(i) + i \text{Im } P_\lambda(i)$, the real and imaginary parts are the signed sums $\text{Re} = \sum_k (-1)^k a_{2k}$, $\text{Im} = \sum_k (-1)^k a_{2k+1}$ where $P_\lambda(t) = \sum_j a_j t^j$, $a_j \geq 0$. Neither is sign-definite, and neither vanishing-set is empty: e.g. at $m = 3$, $\text{Re } P_{(4,2)}(i) = 0$ while $\text{Im } P_{(2,2,1,1)}(i) = 0$. The content of Conjecture 1 is that the two vanishing-sets are *disjoint* (except at $(2,2)$). There is no single Schur-positive function whose support computes $P_\lambda(i)$, because ψ^m is generically irreducible as a t -polynomial $P_\lambda(t)$ (for instance $P_{(5,5)}(t) = t^5 + 10t^3 + 10t^2 + 15t + 6$).

The likely route: a ζ_4 cancellation. The recursion (1) evaluated at $t = i$ reads $P_\lambda(i) = \sum_{\text{horiz}} P_\mu(i) + i \sum_{\text{vert}} P_\mu(i)$, a sum of Gaussian integers. The model $P_\lambda(i) = \sum_\pi i^{\text{vert}(\pi)}$ runs over chains π removing horizontal/vertical 2-strips from λ to $\emptyset$. At $t = -1$ the analogous signed sum $P_\lambda(-1) = \sum_\pi (-1)^{\text{vert}(\pi)} = \chi^\lambda(2^m)$ is resolved by the classical sign-reversing involution whose fixed points are domino tableaux, yielding the 2-core/2-quotient law $\text{ord}_{q=-1} G_\lambda = \lfloor |2\text{-core}|/2 \rfloor$. At $t = i$ one needs a *fourth-order* (ζ_4) cancellation: a grouping of chains into orbits on which $\sum i^{\text{vert}} = 0$, with residual fixed configurations governed by the 4-quotient. This matches the empirical facts that every vanisher is a 4-core and that the staircase 4-core $(4,3,2,1)$ has $G_{(4,3,2,1)}(i) = 24(1+i)^5$, a pure power of the prime $(1+i)$. Constructing this involution and showing its fixed set is nonempty for every $\lambda \neq (2,2)$ is the missing step.

6 Computational verification

All identities were checked by exact symbolic computation (sympy; scripts in `~/projects/scratch/`). Specifically: the reduction Theorem 1 against direct SYT evaluation of $G_\lambda(i)$ for $n \leq 10$ (0 mismatches); the integral representation (3) and the Parseval identity (4) for $m \leq 5$; the conjugation symmetry (2); Propositions 2–4; and Theorem 2 (the vanishing scan to $n = 16$).